

PEIQI YU

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EDUCATION

Carnegie Mellon University

Ph.D. Candidate in Electrical and Computer Engineering

Advised by [Changliu Liu](#)

Pittsburgh, PA

Aug. 2024 – May 2029 (expected)

- Courses: Introduction to Robot Learning (16831), Advanced Control for Robotics (16714), Kinematics for Robot Manipulation (16741), Estimation Detection & Learning (18752), Safe Robotics (16883)

Carnegie Mellon University

M.S. in ECE

Pittsburgh, PA

Aug. 2024 – May. 2026

Tsinghua University

B.Eng. in Automation

Beijing, China

Aug. 2019 – June. 2023

RESEARCH INTERESTS

I work toward self-improving embodied agents for long-horizon robotic manipulation, with a focus on agentic robotics, neuro-symbolic reasoning, and continual learning.

SELECTED PUBLICATIONS

* Equal contribution.

- [1] **Peiqi Yu**, Mosam Dabhi, Shangtao Li, Bowei Li, Laszlo Jeni, and Changliu Liu. “ManiSkillFormer: Demonstration-Free Compositional Manipulation via Geometric Contracts and Agentic Skill Graph”. Submitted to ICRA 2027.
[📄 paper](#) [🌐 Website](#)
- [2] Yiyuan Pan*, Xusheng Luo*, Hanjiang Hu, **Peiqi Yu**, Changliu Liu “Emergent neural automaton policies: Learning symbolic structure from visuomotor trajectories”. **Robotics: Science and Systems (RSS) 2026**.
[📄 arXiv](#) [🌐 Website](#)
- [3] **Peiqi Yu***, Philip Huang*, Chaitanya Chawla*, Guanya Shi, Jiaoyang Li, Changliu Liu. “Autonomous Integration and Improvement of Robotic Assembly using Skill Graph Representations”. **ICRA 2026 Workshop on Multi-Agent Robotic Systems: Real-World Collaboration and Interaction**.
[📄 arXiv](#) [🌐 Website](#)
- [4] Lei Zheng, **Peiqi Yu**, Zengqi Peng, Changliu Liu, Armin Lederer. “Dynamic Neural Koopman Distillation for Real-Time Robot Control Using Diffusion Models”. 2026.
[📄 arXiv](#) [🌐 Website](#)
- [5] Bowei Li*, **Peiqi Yu***, Zhenran Tang, Han Zhou, Yifan Sun, Ruixuan Liu, Changliu Liu. “Nesypack: A neuro-symbolic framework for bimanual logistics packing”. **RSS 2025 Workshop on Benchmarking Robot Manipulation**.
[📄 arXiv](#) [🌐 Website](#)
- [6] **Peiqi Yu***, Abulikemu Abuduweili*, Ruixuan Liu, Changliu Liu. “Robustifying Long-term Human-Robot Collaboration through a Multimodal and Hierarchical Framework”. **IFAC Cyber-Physical & Human Systems 2026**.
[📄 arXiv](#)
- [7] Cheng Ma, **Peiqi Yu**, Jiwen Lu, Jie Zhou. “Recovering realistic details for magnification-arbitrary image super-resolution”. **IEEE Transactions on Image Processing (TIP)**, 2022.

SELECTED RESEARCH EXPERIENCE

Embodied Agent Group, ModelBest Inc.

Research Intern

Beijing, China
Feb. 2024 – July. 2024

Advised by [Zhiyuan Liu](#) | latent action prediction in vision-language models

- Investigated SoTA multimodal and video-based embodied learning methods, including Flamingo, Genie, UniPi, Video PreTraining (VPT), and AVDC, and explored latent-action learning by predicting future visual states and recovering control actions from frame transitions via inverse dynamics.

Intelligence Control Lab, Carnegie Mellon University

Research Intern

Pittsburgh, PA
Sep. 2023 – Dec. 2023

Advised by [Changliu Liu](#) | human-robot collaboration and time-series prediction

- Proposed and developed a multimodal hierarchical HRC framework integrating vision, speech, hierarchical plan prediction, and online user adaptation; deployed it on a Kinova Gen3 for long-horizon toy car assembly task, achieving 91.8% task success, 15.9% faster completion, and 84% user satisfaction. (*The HRC paper*)

Computer Vision Lab, Tsinghua University

Undergraduate Research Intern

Beijing, China
May. 2021 – May 2022

Advised by [Jiwen Lu](#) | 2D Image Super Resolution

- Co-developed Implicit Pixel Flow (IPF), a model-agnostic coordinate-warping framework for photorealistic arbitrary-scale super-resolution. (*The TIP paper*)

SERVICE

Reviewer, IEEE International Conference on Robotics and Automation (ICRA)

2025, 2026

Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2025, 2026

SKILLS

Programming

Python, C/C++, SQL, MATLAB, $\text{\LaTeX}$

Frameworks & Simulators

PyTorch, ROS/ROS2, Isaac Sim, MuJoCo, Gazebo

Robot Platforms

Unitree G1, Galaxea R1Lite, AgiBot G1, Kinova, Yaskawa

HONORS AND AWARDS

First Prize, ICRA WBCD Competition [\[Link\]](#)

2025

Second Prize, Electronics Design Contest, Tsinghua University

2022

Third Prize, Challenge Cup, Tsinghua University

2021